arXiv:2604.23478v3 [cs.CL] 18 Sep 2026

JudgeSense: A Benchmark for Prompt Sensitivity in LLM-as-a-Judge Systems

Rohith Reddy Bellibatlu [*1] Edward Raff[2,3] Wenbin Zhang [1]

[1]Florida International University, Miami, FL

[2]CrowdStrike, New York, USA

[3]University of Maryland Baltimore County, Baltimore, USA

Abstract

Large language models are widely used to judge the output of other language models, yet whether a judge returns the same verdict when the same request is worded differently remains largely unexamined. We study that question across four evaluation tasks and twenty-five judges from six providers. To support the analysis we release JudgeSense, a benchmark of 880 items from human-labelled corpora, each issued under two instructions that differ in wording and not in what they ask, with the complete decision logs. Every score is reported against the judge's own agreement with itself on the identical prompt, so decoding noise is not charged to wording, and the release lets a reader ask the same of any judge not in our roster. Rewording costs agreement on all four tasks, and on two it clears the threshold we declare for a practically meaningful effect; the ordinal task is both the least stable and the one fewest judges are accurate on, and within a single family parameter count does not predict stability. A judge measured inside an agent harness yields a *smaller* estimate than the same judge reached through a direct API call, because its agreement with itself collapses faster than its agreement across wordings.

1 Introduction

Language models now routinely evaluate the output of other language models. A model used this way is a *judge*: it is shown one piece of text, or two, together with an instruction telling it what to decide, and it returns a label from a small fixed set. Throughout this paper "judge" means the model doing the deciding, never the model whose output is being judged. The practice is cheap, fast, and correlates well enough with human preference to have displaced human annotation across much of the evaluation stack (33), and an evaluator earns that role by being reproducible. If two researchers ask the same judge the same question in different words and get different verdicts, the judge is partly measuring the phrasing. The usual validation cannot see that: it asks whether a judge agrees with humans on average, and a judge can correlate well in aggregate while flipping on individual items.

Instability under rephrasing is well documented for models performing ordinary tasks, where formatting, ordering and the choice between synonyms move accuracy substantially (24, 19, 6).

*Corresponding author.

Benchmark and decision logs: `https://huggingface.co/datasets/Rohithreddybc/judgesense-benchmark`. Code, and the scripts behind every table and figure: `https://github.com/rohithreddybc/judgeSense`.

For judges the known failures are positional and length-related: judges favour the first-presented candidate (28, 25), and annotators in the preference votes we draw on chose the longer response in roughly 70% of usable comparisons. Whether the same judge, asked the same question two ways, returns the same answer has been measured far less; Section 2 sets out what has and has not been established.

This paper measures it. We introduce JudgeSense, 880 items across four evaluation tasks, each issued under two instruction templates that differ in wording and not in what they ask. *Factuality* asks whether a statement is factually correct and *coherence* asks how well a summary holds together on a one-to-five scale; *relevance* asks which of two passages better answers a query and *preference* which of two responses is better. The four are independent task families with their own items and their own paraphrase sets, and no item appears in more than one. The quantity we report is the Judge Sensitivity Score,

$$\mathrm{JSS}(j,t) \;=\; \frac{1}{|P_t|} \sum_{(p,p')\in P_t} \delta\big(j(p),\, j(p')\big)\,, \tag{1}$$

where j is a judge, t an evaluation task, and P_t that task's set of *prompt-paraphrase pairs*. $\delta(a,b) = 1$ if $a = b$ and 0 otherwise, evaluated in the task's decision space D_t. A JSS of 1.0 means the judge returned identical decisions on every pair; below 0.8, rewording alone flips its verdict on more than one item in five. This is not accuracy. A judge that is wrong on every item but wrong *identically* under both phrasings scores 1.0, which is intended: we ask whether the instrument is stable and treat whether it is correct as a separate question. One judge measured on one task is a *cell*, the unit every table and figure below is built from.

A prompt-paraphrase pair is two evaluation instructions that ask for the same decision on the same item and differ only in surface wording. One factuality pair, shipped as templates T1 and T3, reads in full:

```
Is this statement factually correct?  Answer YES or NO only.  {text}
Is the following statement accurate?  Answer with one word:  YES or NO. {text}
```

Both request the same two labels in the same direction, the statement substituted into each is byte-identical, and neither asks the negated question. The paraphrase is of the *instruction*, never of the item it carries.

Measuring this well is harder than issuing two prompts and counting disagreements. Three challenges have to be addressed.

i) Difficulty in separating sensitivity to wording from the judge's own noise. A judge disagrees with itself on the identical prompt, so raw disagreement under rewording mixes two sources. The obvious correction, subtracting the one from the other, cannot be trusted everywhere: a judge that already disagrees with itself has little agreement left to lose, so its corrected score is pulled toward zero and noise masquerades as stability. In a pilot we discarded, a misconfigured decoding parameter produced a ceiling low enough to absorb the entire effect we report, and nothing in the output distinguished it from a genuine one. A usable endpoint needs the correction *and* a rule for when the corrected value may be read (Section 4.2).

ii) Difficulty in building a benchmark whose difficulty is not an artifact of its own construction. If a heuristic that ignores the construct can pass, a high score says nothing about judgment. Always answer A. Pick the longer candidate. Pick the passage sharing more words with the query. Yet a benchmark nothing can pass is equally useless, and a construction that suppresses exploitable signal can suppress genuine signal by the same mechanism. Both directions have to be checked, and neither failure shows in labels, provenance or aggregate statistics: every defect we found in our own rebuilds had passed each check that was not looking for it (Section 3).

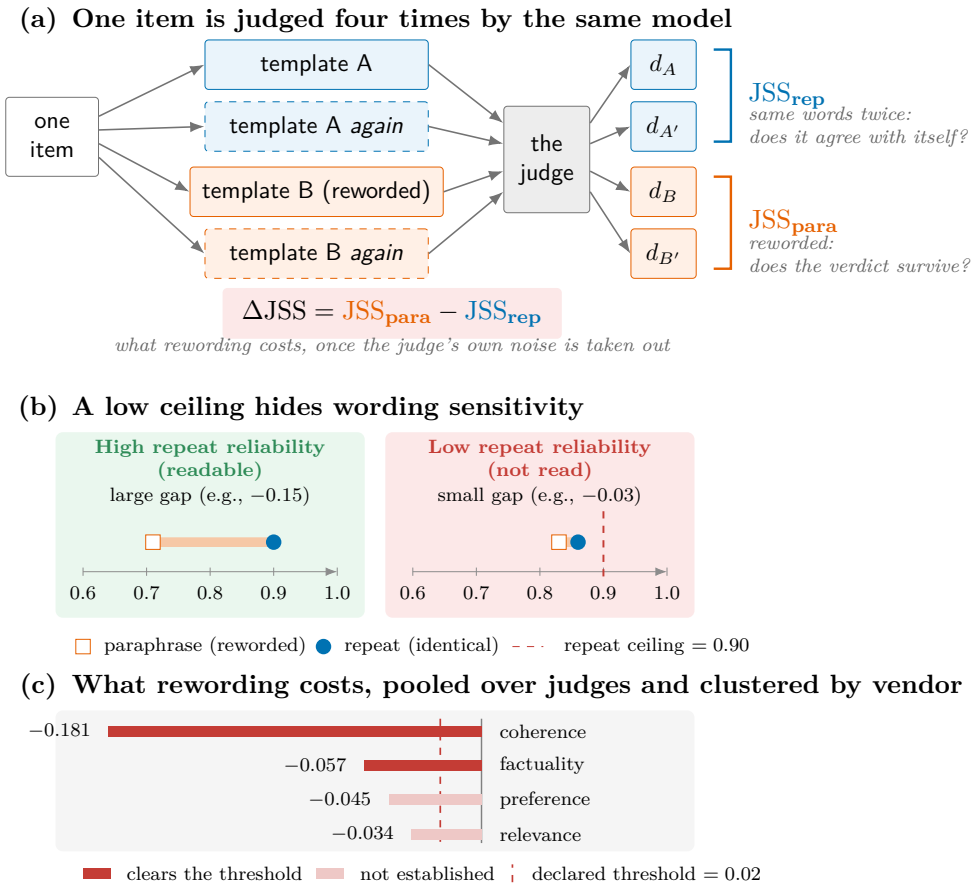


Figure 1: **How one item is judged, and why the endpoint is a difference.** (a) Each item goes to the same judge four times: template A, A again unchanged, a reworded template B, and B again. Blue compares a template with its own repeat, the judge's agreement with itself and so its reproducibility ceiling; orange compares the two wordings. (b) Subtracting one from the other charges decoding noise to the ceiling, so a judge that already disagrees with itself has less left to lose and shows a *smaller* gap, which is why cells below 0.90 are reported but not read. (c) The answer, pooled over judges and clustered by vendor (Table 3). Takeaway: rewording moves a judge's verdict by more than its own decoding noise does, most of all when the judge is asked for a rating rather than a choice.

iii) Difficulty in drawing a claim from judges that are not independent draws. Checkpoints from one vendor share pretraining corpora, tuning recipes and decoding defaults, so a sensitivity learned once reappears across siblings and is counted once per sibling. Treating them as separate evidence is what produced the claim this work has since retracted. Counting them honestly costs precision, and the cost is large enough to change conclusions (Section 4.3).

This paper is a correction as well as a contribution. An earlier version of this benchmark (3) was withdrawn after review. Its 75 items were hardcoded in source while being described as benchmark-sourced; it presented the correct candidate first in every pairwise item, so a judge that always answered "A" was indistinguishable from a perfect one; and one factuality template inverted the answer polarity, inflating measured disagreement uniformly across every judge. We rebuilt from real corpora with per-item provenance, and the rebuild had further defects of the same class: a

template assignment that functioned as an answer key, candidate responses carrying contradictory gold labels, and a length-balancing step that installed a difficulty confound in place of the verbosity one. We report those too. None is exotic, and each survived every check that was not specifically looking for it.

Contributions. To tackle the above challenges, this paper introduces JudgeSense. We state first what is not ours: the agreement rate in Equation 1 is what Elazar et al. (6) use for consistency in pretrained models, what Arabzadeh and Clarke (2) compute against relevance labels, and what Zhuo et al. (34) express as a variance, while Ahn and Yin (1) distinguish inconsistency under repetition from inconsistency under rewording. No single prior source covers all of what follows; we borrow the measure and the distinction, and supply the rest.

- **An endpoint measured against the judge's own noise (i).** We subtract repeat agreement from paraphrase agreement, scoring each judge against its own reproducibility ceiling, and decline to read any cell whose ceiling is too low to carry the comparison.
- **A benchmark whose attack surface is measured, not asserted (ii).** 880 items from four human-labelled corpora with per-item provenance, released with the heuristic baselines that would beat it, their intervals and support counts, and a declared threshold the best judge must clear for a task to count as discriminating.
- **Inference that does not count siblings twice (iii).** The confirmatory contrast clusters judges by the vendor that trained the checkpoint. Two of our four claims do not survive it, and we say so.
- **Refusal treated as a measurement.** Provider-flagged declines are recorded apart from unparseable output: alike once parsing has failed, but different facts about a judge.
- **Evidence that the ordinal task is the least stable.** Coherence carries the largest pooled effect of the four, roughly three times the next largest, still the largest when its scale is collapsed to binary, and the largest loss for fourteen of the sixteen judges whose four cells are all readable. It is also where judges differ most from each other: the least stable task and the least agreed-upon one at once.

2 Related work

LLM judges and their validation. Zheng et al. (33) established the practice at scale through MT-Bench and Chatbot Arena, and holistic evaluation efforts adopted it (16). Validation is almost always correlational: a judge is trusted when its scores agree with human ratings in aggregate (4, 18). Chiang and Lee (4) find that agreement broadly positive but task-dependent, which prefigures a result below. What correlational validation cannot say is whether rerunning an evaluation under differently worded instructions will reproduce it.

Known judge biases. Wang et al. (28) show position bias, the preference for whichever candidate is presented first, to be strong enough to require explicit correction. Shi et al. (25) extend that across twelve judges and more than 100,000 evaluation instances, and find it a stable, model-specific property that bites hardest when the two candidates are close in quality. Length is the other documented confound (28). Both describe what judges prefer, and Section 3 treats them as constraints on construction. Neither reports how a verdict moves when the instruction itself is reworded.

Prompt sensitivity in language models generally. Formatting choices alone swing accuracy by as much as 76 points on some benchmarks (24), paraphrasing task instructions alters model rankings (19), and consistency under rephrasing has been treated as a first-class property of pretrained models (6). The sensitivity has itself been benchmarked (34, 22, 23): Razavi et al. (22) report that current methods cannot anticipate which prompts will trip a model up, which argues for measuring it directly, and Romanou et al. (23) attribute up to half of a model's performance variance to semantics-preserving perturbations. Wei et al. (29) study prompt-template effects on judge reliability, restricted to alignment tasks and under a different metric.

Separating the two inconsistencies is not our idea. Ahn and Yin (1) name them: inconsistency under repetition, a property of decoding, and inconsistency under rewording, a property of the prompt. We subtract the first from the second, report the difference, and treat the first as a precondition for reading the second at all. Weng et al. (30) apply the same correction elsewhere, counting verdict flips on safety rubrics only where they exceed a judge's own baseline jitter.

All of this concerns a model being tested. A judge does not generate an answer; it returns a label from a small finite set, so its sensitivity is a question about decision stability, not output fluency, and these measurements do not transfer to the model doing the testing.

Judging judges. Thakur et al. (26) assess thirteen judges on a clean alignment task and document sensitivity to prompt complexity, length and other surface features alongside a general tendency toward leniency; Li et al. (14) survey the area broadly. The last two years have made judge reliability a subject in its own right. Li et al. (15) find reliability highly sensitive to the choice of judge prompt template; Norman et al. (20) evaluate twenty-one judges and show that exact-match agreement overstates discrimination once corrected for chance, which is why we report Cohen's κ beside every raw figure; and Feng et al. (8) score judges on self-consistency without human annotation, as we do. Closest in method is Yagubyan (31), who repeat identical evaluations fifty times per question and report that semantically equivalent templates change the majority outcome in a quarter of cases, on two judges from one provider.

Closest in question is Arabzadeh and Clarke (2), who compare human- and LLM-authored prompt variants for relevance judgment (90 prompts, three judge models) against TREC labels using Cohen's κ, the agreement rate discounted for what two raters would reach by chance. The difference is the referent: that work tracks how rewording moves a judge's *accuracy* against human gold, while JSS never consults the gold label and tracks how rewording moves a judge's *agreement with itself*. The two diverge when a judge flips verdicts in compensating directions, which our measurement sees and an accuracy-referenced one does not.

Why not simply evaluate under many prompts? Hua et al. (11) dispute the premise, arguing that much reported prompt sensitivity is an artifact of log-likelihood scoring and rigid answer matching, and that an LLM judge in their place cuts variance across templates. We read that as motivating this paper: their comparison is of ways to score a model, ours is of the scorer. If a judge absorbs the benchmark's prompt sensitivity, how much the judge itself carries is the question that remains, and it is answered by measuring rather than by assuming the substitution is free. Our parsing is strict and our decision spaces small and fixed, so the leniency their critique targets cannot inflate our estimate.

Mizrahi et al. (19) advocate multi-prompt evaluation as a methodological default. It is good advice for the evaluation being run, but it must be re-run for every new judge over a prompt distribution chosen for that study, whereas a fixed paraphrase set yields a number computed once and portable anywhere. Variance-based characterisations (24, 34) need gold labels, since they

measure how accuracy moves across prompt variants; ours needs none, so it applies where no gold labels exist, which is the usual situation. Ensemble judging (14) manages sensitivity by combining judges, and says nothing about whether any individual judge is stable enough to use alone.

In contrast to existing work, this paper measures the stability of the judge itself, against its own noise rather than against a gold label, across twenty-five judges from six providers, and reports the result under an independence assumption weak enough that two of its four claims do not survive.

Sources. The items come from existing human-labelled corpora: TruthfulQA for factual accuracy (17), SummEval for expert coherence ratings (7), BEIR's TREC-COVID split for graded relevance judgements (27), and MT-Bench's human preference votes (33). No label in this benchmark is assigned or inferred by us. The judges are drawn from current instruction-tuned and reasoning-tuned families (21, 9, 10, 32, 12, 5).

3 The JudgeSense benchmark

JudgeSense measures whether an LLM judge returns the same verdict when the same evaluation request is worded differently. A judge whose verdict depends on which of two equivalent phrasings it received is not usable as a measuring instrument, however accurate it looks on average.

3.1 Tasks and sources

The benchmark covers four evaluation tasks, each drawn from an existing human-labelled corpus. No item is authored by us and no label is assigned or inferred by us; every label is the one the source corpus already carried.

Factuality (250 items, TruthfulQA). A statement is judged accurate or inaccurate. Items whose question is ill-posed, and whose only defensible answer is a refusal to answer, are excluded at load time.

Coherence (250 items, SummEval). A summary is rated 1–5 for coherence against expert annotations. Summaries with identical text but conflicting expert labels are dropped, not merged.

Relevance (250 items, 500 rows, BEIR TREC-COVID). Given a topic, the judge picks the more relevant of two passages. *Both* candidates carry a human relevance grade: the positive is graded fully relevant (2) and the distractor is one a human assessor explicitly rejected (0). Partially relevant documents (grade 1) are used for neither role.

Preference (130 items, 260 rows, MT-Bench human judgments). The judge picks the response humans preferred. Only comparisons with at least two annotator votes and a clear majority are used.

The relevance construction departs from the usual practice of drawing distractors from a positives-only collection. There, a negative can only be a document *absent* from an incomplete judgment set, so a topically similar distractor may in fact be relevant but unjudged, and the item penalises a defensible answer. TREC-COVID carries tens of thousands of explicit non-relevant assessments, so both candidates here are human-labelled and the distractor is one a human affirmatively rejected.

Every shipped item records its source dataset, split, per-record identifier and source configuration, so any item can be traced to the row it came from. When a source is unavailable the loader raises; there is no fallback path that could quietly generate an item in its place.

3.2 Paraphrase design

Each item is issued under two of five instruction templates, intended to differ in wording and not in what they ask; all twenty are printed in full in Appendix E. That intent is a construct assumption: if it fails, a judge answering both arms correctly is scored as unstable, and the result is semantic drift, not prompt sensitivity.

We audit the released files, not the generator, recovering each template from the shipped prompts and the payload beside them. The audit is offline, deterministic, runs in continuous integration, and enforces four properties. *Label space*: both templates in a pair request the same answer set ({YES, NO} for factuality, $\{1, \ldots, 5\}$ for coherence, {A, B} for the pairwise tasks). *Polarity*: no answer token is bound to opposite conclusions, and no template asks the inverted question. *Construct*: every template draws its vocabulary from its own task. *Non-triviality*: no pair is a near-duplicate. All four hold on all twenty shipped templates.

The limits of the audit. Those properties are structural. None shows that two differently worded instructions ask for the same *judgement*, the assumption the benchmark rests on. That assumption is smaller than the item count suggests, and its size is what makes it checkable: equivalence is a property of the instruction pair, not of the item, and within an item both arms carry byte-identical content, so no particular passage can make one arm ask a different question. The 880 items draw on ten instruction pairs per task, forty in all, built from the twenty templates printed in Appendix E, so a reader can audit the assumption directly against the four decision spaces.

We did not have those forty pairs scored by an independent judge, human or model, for this release. The manual records from the first version are keyed to identifiers no item here uses. A model-based check returned unanimous agreement, but its validators consulted the generator and the equivalence tests instead of judging the pairs as shown; that is circular, and we rest nothing on it.

Two further properties we measure and report but cannot enforce. Templates place the payload at different points: in relevance, T1 opens the query at 46.4% of the template against T3 at 9.0%, a within-pair divergence of 37.4 percentage points; and reading difficulty varies by up to 3.7 Flesch–Kincaid grade levels within a pair. Either could make one arm intrinsically harder. Three residuals remain open: the audit groups surface forms into one construct by an authored mapping, so whether "factually correct" and "accurate" are synonyms for a given judge is an assumption; implicit thresholds may shift between phrasings even when the construct does not; and tokenisation or pretraining frequency may favour one phrasing for one model. Semantic drift is reduced as an alternative explanation, not eliminated.

A limitation of the assignment. Each item is issued under exactly one of the ten unordered template pairs, balanced across pairs, so template identity nests perfectly inside item identity: the items behind T_i and T_j are disjoint, and any apparent difference between two templates is equally a difference between the items they were shown. Claims of the form "template A is more stable than template B" are not identifiable here, and we make none. The within-item contrast the design was built for is unaffected, since the two arms of a pair differ only in wording with the item held fixed. Appendix A records the polarity defect this design corrects.

3.3 Position control

Both pairwise tasks present every item twice, once in each candidate ordering. The correct answer therefore sits at position A in exactly half the rows, and a judge that ignores content and always

answers "A" scores 0.500, not 1.000.

This corrects an earlier version in which the correct candidate always appeared first, where most judges scored a degenerate JSS = 1.000 that reflected the layout and no judgment at all.

3.4 Release and integrity

The dataset is released with Croissant metadata, per-item provenance, and a content hash for each split that excludes retrieval timestamps, so a rebuild can be verified as identical in content without being byte-identical in metadata. A 32-check audit gate (eight checks over each of four splits) runs in continuous integration and fails the build on duplicate content, unbacked provenance claims, label degeneracy, insufficient clustering support, contradictory ground truth, or a length shortcut exceeding its bound. A check that cannot read the data it is meant to inspect fails; it does not pass. A check that silently measures nothing therefore cannot be mistaken for one that found nothing wrong.

4 Measuring sensitivity

This section settles three things before any number is reported: what is measured on a single cell, how the cells are pooled into a claim about a task, and which cells are read at all. The first is the endpoint, ΔJSS, defined in Section 4.2 as the gap between a judge's agreement across two wordings and its agreement with itself. The second is a choice about independence, since sibling checkpoints from one vendor are not separate evidence (Section 4.3). The third is a threshold on the repeat arm, below which a cell is reported and not read (Section 4.4); that bar conditions on a term of the quantity it filters, so the contrast is reported with and without it.

4.1 Judge Sensitivity Score

Equation 1 gives the endpoint's first term: the rate at which a judge returns the same decision under two equivalent templates. It never consults the ground-truth label, so accuracy is reported separately. On the pairwise tasks the companion measure is *position-corrected* accuracy, whether the judge selected the side carrying the ground truth given that the swap design places it at position A in exactly half the rows; on the pointwise tasks it is ordinary accuracy against the source label. Raw agreement is also inflated when a judge's output distribution is compressed, since a judge answering "YES" almost always agrees with itself, so alongside every raw figure we give the chance-corrected form (Cohen's κ, quadratic-weighted for the ordinal coherence task).

As Section 1 states, the agreement rate itself is borrowed. What is new here is the benchmark, the systematic application to the model doing the judging, and the endpoint defined next.

4.2 The repeat baseline, and why it is the endpoint

Raw JSS below 1.0 does not by itself demonstrate sensitivity to *wording*. A judge sampling at nonzero temperature, or served through a non-deterministic backend, disagrees with itself on the identical prompt, and raw JSS attributes that decoding noise to paraphrasing. We separate the two by issuing one template a second time, unchanged. The resulting $\mathrm{JSS}_{\mathrm{rep}}$ is the judge's reproducibility ceiling on that task, and the primary endpoint is the difference

$$\Delta\mathrm{JSS} = \mathrm{JSS}_{\mathrm{para}} - \mathrm{JSS}_{\mathrm{rep}},$$

which is negative when rewording costs more stability than resampling does.

A pre-registration is worth only what is true of it, and ours is partly post-hoc. Commit `b1b507a`, before the full sweep, fixed the endpoint, the estimator, the item-level clustering unit, and one decision rule: an effect is present when the interval excludes zero. Those four preceded the sweep though not the first API call, since a five-row-per-cell smoke wave had already exercised the harness against three judges; it produced no reportable estimate and none of its output enters the results. Commit `82ea522`, some seventy minutes after the first full judge's results were committed, added the support floors, the multiplicity plan and the discrimination thresholds. Those are *not* pre-registered, and two of the four discrimination thresholds were set while the corresponding accuracies were already visible in the repository, which disqualifies them as confirmatory tests; they are reported as post-hoc descriptive checks. One correction is recorded in the plan itself: the direction of subtraction in the endpoint's prose was stated backwards relative to the estimator, which had been implemented and tested beforehand.

The multiplicity plan added at `82ea522` is the one this paper applies: a pooled contrast per task with Holm correction, per-cell intervals demoted to exploratory under Benjamini–Hochberg control at a 10% false discovery rate, and a reported minimum detectable effect. No judge is ranked against another on the strength of an unadjusted cell. Because the plan was written after the first judge's results were committed, the confirmatory and exploratory labels are themselves declared rather than pre-registered, and we carry that qualification wherever they are used.

4.3 Pooling across judges

The confirmatory claims pool judges within a task, and the estimator is the unweighted mean of the judge-level ΔJSS values for that task's readable cells. Weighting by precision would let the judges with the narrowest intervals decide a cross-judge claim, which is not what we are asking.

The sampling unit is the judge, and judges are not independent: twenty-five checkpoints come from eight vendors, one of which supplies seven. Sibling checkpoints share pretraining corpora, instruction-tuning recipes and decoding defaults, so a sensitivity learned once reappears across the siblings and would be counted once per sibling. That is the error we attribute to the superseded version of this work, whose retracted claim rested on three judges from one vendor. Kohli (13) put a number on the same effect for judge panels: across nine frontier judges from seven families, correlated errors leave roughly two votes' worth of independent information, so counting judges overstates the evidence they jointly carry. We therefore report a cluster-robust standard error with the vendor as the cluster, using the CR1 correction and Student's t on $G - 1$ degrees of freedom instead of a normal approximation, which with eight clusters is optimistic. Both the point-null test and the equivalence-style test against the declared threshold are Holm-corrected over their own family of four, and the unclustered standard error is reported alongside so the cost of the assumption is visible.

4.4 The readability bar, and what it costs

We decline to read any cell whose repeat ceiling falls below 0.90. The bar is declared rather than pre-registered: it was fixed after the pilot described in Section 5, in which a misconfigured decoding parameter produced a ceiling of 0.864 and absorbed the entire effect.

It also conditions on a term of the quantity it filters, since ΔJSS subtracts $\mathrm{JSS}_{\mathrm{rep}}$ and the bar is a threshold on $\mathrm{JSS}_{\mathrm{rep}}$, so retaining only high-ceiling cells is not a neutral quality gate. We report the contrast both ways. Dropping the bar moves coherence from -0.181 to -0.158 and relevance from -0.034 to -0.039; factuality is unchanged at -0.057 and preference moves from -0.045 to -0.042. The bar flatters the largest effect and penalises the smallest, the direction that favours our

own headline. Every pooled conclusion in Section 5 holds under both.

We also report a minimum detectable effect per cell: the smallest $|\Delta\text{JSS}|$ that cell could have told apart from zero at its own support, computed as $(z_{0.975} + z_{0.80})$ times the bootstrap standard error of its estimate, the conventional two-sided calculation at 80% power. A cell whose estimate falls below its own minimum detectable effect is uninformative rather than null, and thirty-six of the eighty-seven readable cells are in that position.

4.5 Uncertainty and decoding budget

All intervals are 95% cluster bootstrap intervals over 2,000 resamples with the item as the resampling unit. The design forces that choice: a pairwise item contributes two rows, one per candidate ordering, and its repeat arm nests inside it, so resampling rows independently treats correlated observations as independent and understates uncertainty. The clustering unit is a mandatory argument in our implementation, with no default, so it cannot be omitted by accident. An effect is reported as present for a judge–task cell when the interval for ΔJSS excludes zero, and the direction is read from the sign.

Every judge is run with an identical output token limit of 1,024 tokens. A class-dependent limit, tight for instruction-tuned judges and generous for reasoning-tuned ones, would confound judge class with inference configuration, so any difference between the classes would admit an explanation having nothing to do with the models; it would also truncate real responses, since a judge that emits any preamble before its verdict exhausts a tight cap mid-sentence and the fragment is retained as its answer. Because a token limit is a ceiling and not a target, the matched budget costs nothing for judges that emit a single-token answer, and more only for the verbose non-compliers whose calls were previously truncated into noise.

We quote no rate for any of this. Every committed run was made under the matched policy, so we hold no measurement of the cap-termination rate under the class-dependent one, and quantifying the budget's effect would require a subsample run under that configuration. We did not run it, and we make no claim about the effect's size; the case for the matched budget rests on the confound being structural.

5 Results

We report twenty-five judges across six providers on the four tasks, giving ninety-eight judge–task cells with an endpoint. Two cells report none: the Sonnet and Opus preference splits stopped below the declared support floor when the API balance was exhausted mid-run, and Section 5.9 states what that costs. Every figure below is produced by `scripts/regenerate_results.py` from the committed raw outputs; none is transcribed.

Table 1: **Paraphrase sensitivity per judge and task.** $\Delta\text{JSS} = \text{JSS}_{\text{para}} - \text{JSS}_{\text{rep}}$ is the pre-registered endpoint, negative when rewording costs more agreement than re-issuing the identical prompt does. The bar draws that estimate from a zero line, leftward for a cost, with the hairline at the declared 0.02 threshold; bands group one judge's four tasks. Intervals are 95% item-clustered bootstrap over 2,000 resamples, and cells below the support floor of 100 clusters report no endpoint. r is the refusal rate and m the malformed-output rate over the arms the judge attempted. The first two columns do not subtract to the printed delta wherever a cell carries refusals or a swapped-ordering imbalance, because the delta uses the canonical ordering only. Takeaway: rewording costs agreement almost everywhere, and coherence is the largest loss for nearly every judge and the column they disagree on most.

Judge	Task	n	JSS	JSS_{rep}	*effect*	ΔJSS [95% CI]	r	m
claude-haiku	factuality	250	0.956	1.000		−0.044 [-0.072, -0.020]	0.000	0.000
	coherence	250	0.792	0.998		−0.206 [-0.258, -0.154]	0.000	0.000
	relevance	500	0.930	0.990		−0.058 [-0.086, -0.032]	0.000	0.010
	preference	260	0.896	0.965		−0.081 [-0.127, -0.038]	0.000	0.031
claude-opus-4-7	factuality	250	0.960	0.990		−0.030 [-0.056, -0.006]	0.000	0.000
	coherence	250	0.736	0.916		−0.180 [-0.236, -0.128]	0.000	0.000
	relevance	500	0.964	0.998		−0.036 [-0.063, -0.013]	0.051	0.000
	preference	170	0.963	—		*support below floor*	0.000	0.056
claude-sonnet	factuality	250	0.952	0.990		−0.038 [-0.062, -0.018]	0.000	0.008
	coherence	250	0.828	0.990		−0.162 [-0.210, -0.118]	0.000	0.000
	relevance	500	0.584	0.789		−0.158 [-0.206, -0.110]	0.247	0.224
	preference	44	0.559	—		*support below floor*	0.000	0.432
deepseek-v4-flash	factuality	250	0.888	0.908		−0.020 [-0.056, 0.016]	0.000	0.010
	coherence	250	0.616	0.670		−0.054 [-0.116, 0.008]	0.000	0.000
	relevance	500	0.944	0.962		−0.006 [-0.026, 0.012]	0.000	0.004
	preference	260	0.911	0.923		−0.023 [-0.058, 0.012]	0.000	0.025
deepseek-v4-flash-ds	factuality	250	0.888	0.900		−0.012 [-0.046, 0.022]	0.000	0.004
	coherence	250	0.628	0.630		−0.002 [-0.062, 0.060]	0.000	0.004
	relevance	500	0.946	0.952		−0.008 [-0.028, 0.012]	0.000	0.016
	preference	260	0.892	0.900		+0.000 [-0.038, 0.038]	0.000	0.050
deepseek-v4-pro	factuality	250	0.904	0.922		−0.018 [-0.046, 0.010]	0.000	0.042
	coherence	250	0.720	0.742		−0.022 [-0.076, 0.032]	0.000	0.004
	relevance	500	0.960	0.968		+0.008 [-0.004, 0.022]	0.000	0.014
	preference	260	0.923	0.896		+0.019 [-0.012, 0.058]	0.000	0.048
gemini-3.7-flash	factuality	250	0.976	0.992		−0.016 [-0.032, -0.004]	0.000	0.000
	coherence	250	0.840	0.902		−0.062 [-0.102, -0.022]	0.000	0.000
	relevance	500	0.982	0.992		−0.012 [-0.026, 0.000]	0.000	0.000
	preference	260	0.969	0.988		−0.019 [-0.046, 0.000]	0.000	0.000
gemini-flash	factuality	250	0.896	0.996		−0.100 [-0.136, -0.064]	0.000	0.000
	coherence	250	0.688	0.986		−0.298 [-0.358, -0.242]	0.000	0.000
	relevance	500	0.878	0.950		−0.058 [-0.084, -0.034]	0.000	0.056
	preference	260	0.935	0.996		−0.027 [-0.058, -0.004]	0.000	0.002
gemma-4-31b	factuality	250	0.980	0.992		−0.012 [-0.032, 0.004]	0.000	0.000

continued overleaf

Table 1, continued from the previous page.

Judge	Task	n	JSS	JSS_{rep}	*effect*	ΔJSS [95% CI]	r	m
	coherence	250	0.848	0.958		−0.110 [-0.156, -0.070]	0.000	0.000
	relevance	500	0.866	0.904		−0.040 [-0.062, -0.020]	0.000	0.092
	preference	260	0.996	0.996		−0.004 [-0.023, 0.012]	0.000	0.000
glm-5.2	factuality	250	0.880	0.910		−0.030 [-0.056, -0.004]	0.000	0.048
	coherence	250	0.616	0.670		−0.054 [-0.118, 0.012]	0.000	0.002
	relevance	500	0.950	0.970		+0.002 [-0.018, 0.022]	0.000	0.007
	preference	260	0.873	0.881		−0.012 [-0.042, 0.019]	0.000	0.067
kimi-k3	factuality	250	0.948	0.992		−0.044 [-0.072, -0.018]	0.000	0.000
	coherence	250	0.832	0.904		−0.072 [-0.118, -0.028]	0.000	0.000
	relevance	500	0.972	0.994		−0.014 [-0.030, -0.002]	0.000	0.000
	preference	260	0.965	0.992		−0.015 [-0.042, 0.004]	0.000	0.000
llama-3.3-70b	factuality	250	0.952	1.000		−0.048 [-0.076, -0.024]	0.000	0.000
	coherence	250	0.772	0.992		−0.220 [-0.274, -0.170]	0.000	0.000
	relevance	500	0.890	0.926		−0.046 [-0.072, -0.022]	0.000	0.058
	preference	260	0.965	1.000		−0.015 [-0.038, 0.000]	0.000	0.004
llama-4-maverick	factuality	250	0.916	0.996		−0.080 [-0.116, -0.048]	0.000	0.000
	coherence	250	0.764	0.986		−0.222 [-0.274, -0.170]	0.000	0.000
	relevance	500	0.962	0.998		−0.022 [-0.044, -0.006]	0.000	0.006
	preference	260	0.942	0.988		−0.058 [-0.104, -0.019]	0.000	0.002
llama-4-scout	factuality	250	0.892	0.994		−0.102 [-0.142, -0.066]	0.000	0.000
	coherence	250	0.644	0.974		−0.330 [-0.390, -0.272]	0.000	0.000
	relevance	500	0.838	0.974		−0.070 [-0.104, -0.040]	0.000	0.035
	preference	260	0.846	0.938		−0.131 [-0.192, -0.077]	0.000	0.027
llama3-8b	factuality	250	0.884	0.998		−0.114 [-0.156, -0.074]	0.000	0.000
	coherence	250	0.668	1.000		−0.332 [-0.392, -0.272]	0.000	0.000
	relevance	500	0.896	0.992		−0.072 [-0.106, -0.044]	0.000	0.005
	preference	260	0.911	0.981		−0.058 [-0.100, -0.023]	0.000	0.015
magistral-small	factuality	250	0.804	0.944		−0.140 [-0.190, -0.094]	0.000	0.000
	coherence	250	0.820	0.944		−0.124 [-0.170, -0.080]	0.000	0.000
	relevance	500	0.880	0.936		−0.088 [-0.130, -0.048]	0.000	0.002
	preference	260	0.846	0.935		−0.112 [-0.173, -0.054]	0.000	0.000
mistral-medium	factuality	250	0.888	0.980		−0.092 [-0.130, -0.056]	0.000	0.000
	coherence	250	0.892	0.978		−0.086 [-0.124, -0.052]	0.000	0.000
	relevance	500	0.946	0.976		−0.060 [-0.094, -0.028]	0.000	0.000
	preference	260	0.927	0.985		−0.069 [-0.112, -0.031]	0.000	0.004
mistral-small	factuality	250	0.816	0.930		−0.114 [-0.162, -0.066]	0.000	0.000
	coherence	250	0.804	0.930		−0.126 [-0.172, -0.080]	0.000	0.000
	relevance	500	0.840	0.922		−0.114 [-0.164, -0.068]	0.000	0.001
	preference	260	0.839	0.923		−0.100 [-0.173, -0.035]	0.000	0.000

continued overleaf

Table 1, continued from the previous page.

Judge	Task	n	JSS	JSS_{rep}	*effect*	ΔJSS [95% CI]	r	m
qwen	factuality	250	0.908	1.000		−0.092 [-0.128, -0.056]	0.000	0.000
	coherence	250	0.860	0.988		−0.128 [-0.170, -0.086]	0.000	0.000
	relevance	500	0.940	0.974		−0.026 [-0.046, -0.008]	0.000	0.029
	preference	260	0.965	1.000		−0.031 [-0.062, -0.008]	0.000	0.000
qwen-3.6-flash	factuality	250	0.932	0.970		−0.038 [-0.068, -0.012]	0.000	0.000
	coherence	250	0.760	0.912		−0.152 [-0.202, -0.104]	0.000	0.000
	relevance	500	0.944	0.986		+0.002 [-0.014, 0.018]	0.000	0.004
	preference	260	0.973	0.992		−0.031 [-0.065, -0.004]	0.000	0.002
qwen3-14b	factuality	250	0.888	0.930		−0.042 [-0.076, -0.010]	0.000	0.006
	coherence	250	0.576	0.780		−0.204 [-0.264, -0.144]	0.000	0.000
	relevance	500	0.932	0.960		−0.004 [-0.020, 0.012]	0.000	0.028
	preference	260	0.869	0.912		−0.050 [-0.085, -0.019]	0.000	0.064
qwen3-32b	factuality	250	0.952	0.996		−0.044 [-0.072, -0.020]	0.000	0.000
	coherence	250	0.704	0.976		−0.272 [-0.328, -0.216]	0.000	0.000
	relevance	500	0.952	0.992		−0.036 [-0.062, -0.014]	0.000	0.007
	preference	260	0.954	0.996		−0.027 [-0.062, 0.000]	0.000	0.000
qwen3-8b	factuality	250	0.852	0.912		−0.060 [-0.098, -0.020]	0.000	0.008
	coherence	250	0.560	0.728		−0.168 [-0.232, -0.106]	0.000	0.008
	relevance	500	0.870	0.908		−0.032 [-0.050, -0.014]	0.000	0.082
	preference	260	0.769	0.827		−0.065 [-0.115, -0.023]	0.000	0.112
qwen3.7-flash	factuality	250	0.960	0.990		−0.030 [-0.054, -0.008]	0.000	0.000
	coherence	250	0.788	0.936		−0.148 [-0.196, -0.102]	0.000	0.000
	relevance	500	0.954	0.986		−0.010 [-0.026, 0.004]	0.000	0.001
	preference	260	0.985	1.000		−0.015 [-0.038, 0.000]	0.000	0.000
qwen3.8-27b-hf	factuality	250	0.792	0.852		−0.060 [-0.090, -0.030]	0.000	0.114
	coherence	250	0.760	0.968		−0.208 [-0.260, -0.156]	0.000	0.002
	relevance	500	0.946	0.980		−0.024 [-0.046, -0.004]	0.000	0.011
	preference	260	0.911	0.938		−0.031 [-0.065, 0.000]	0.000	0.038

5.1 Headline findings

Six findings, ordered by how much each bears on choosing a judge. The analyses behind them follow in Sections 5.3 onward.

Finding 1: Rewording costs agreement on all four tasks, and on two of them the cost clears the threshold once judges are clustered by vendor. The pooled ΔJSS is negative on every task, from coherence (−0.181) down to relevance (−0.034). Clustering by vendor and correcting over four tests, coherence and factuality clear the 0.02 threshold; preference and relevance do not (Table 3). Unclustered standard errors are 1.37 to 1.63 times smaller and would have cleared all four, but seven of the twenty-five judges come from one vendor, and sibling checkpoints are not independent draws. Per cell, sixty-eight of the eighty-seven readable cells are significant under

Benjamini–Hochberg at a 10% false discovery rate, which is the exploratory reading and carries no confirmatory claim (Section 4.2).

Finding 2: Coherence carries the largest effect and is the task fewest judges are accurate on. It has both the largest pooled loss, three times factuality, and the weakest discrimination: six of twenty-five judges clear its declared accuracy threshold of 0.40, against twenty-five of twenty-five on relevance at 0.70 (Tables 3 and 8). Quadratic-weighted κ, which credits a near miss, spans 0.579 to 0.941 over the sixteen fully readable judges, so they may agree on the ordering while exact-match accuracy punishes placement on a five-point scale. Either way, a single coherence score is unreliable.

Finding 3: Most of coherence's excess is the width of its five-point scale, not the task. Collapsing coherence to a binary label at four or above removes 58.6% of the effect, averaged over judges (Table 5). What remains still leads the four tasks, at about 1.3 times factuality, not three. A five-point scale offers more adjacent categories for a paraphrase to shift a verdict into. The control is not decisive, since a different cut point would remove a different share.

Finding 4: A low repeat ceiling hides wording sensitivity rather than exposing it. Repeat agreement ranges from 0.630 to 1.000 across the ninety-eight cells, and thirty-three sit at 0.99 or above (Figure 3). Because ΔJSS subtracts repeat agreement from paraphrase agreement, decoding noise is charged to the ceiling: a judge that disagrees with itself on identical input has less room left to lose on reworded input, so its estimate is pulled toward zero. The coherence extremes fit that account, `llama3-8b` losing 0.332 from a ceiling of 1.000 against `gemini-3.7-flash`'s 0.062 from 0.902. Eleven cells whose ceiling falls below 0.90, six of them coherence, are therefore reported but not read, leaving eighty-seven. Read the ceiling column first, and treat a near-zero estimate above a low ceiling as uninformative rather than as stability.

Finding 5: Parameter count does not predict stability, and recency is not the story the earlier version told. Within the `qwen-3` ladder, size does not order ΔJSS on any task, and on relevance the middle model is the most stable (Table 7). Across families the coherence extremes are what a capability story would predict, but fourteen of the sixteen fully readable judges lose most on coherence whatever their size. The ladder is incomplete below 8B. Recency is not controlled at all, and what the roster shows descriptively runs against the earlier version's claim that the newest models are the least consistent: ranking the sixteen fully readable judges by mean endpoint, the newest Gemini is the most stable of them and the two newest Qwen checkpoints rank fourth and fifth, while the older `gemini-flash` and `llama3-8b` sit near the bottom. `llama-4-scout` is the counterexample, recent and least stable of the sixteen. We report the pattern and claim nothing from it, because generation is confounded with family, size and training recipe across vendors.

Finding 6: The same judge measured inside an agent harness yields a smaller estimate. Re-measuring Haiku 4.5 through an agent harness instead of a direct API call contracts ΔJSS by 40 to 86% (Table 9). The judge has not become more stable. Its ceiling falls faster under the harness than paraphrase agreement does, and the subtraction in Finding 4 turns that fall into a smaller endpoint. Harness ceilings are far higher on factuality than on coherence (Table 10), so the collapse is task-dependent, not model-dependent. One judge and one harness are a narrow base, but what follows for practice is that an estimate of wording sensitivity does not travel across measurement surfaces: a number obtained from the API should be re-measured inside the harness the judge will actually run in.

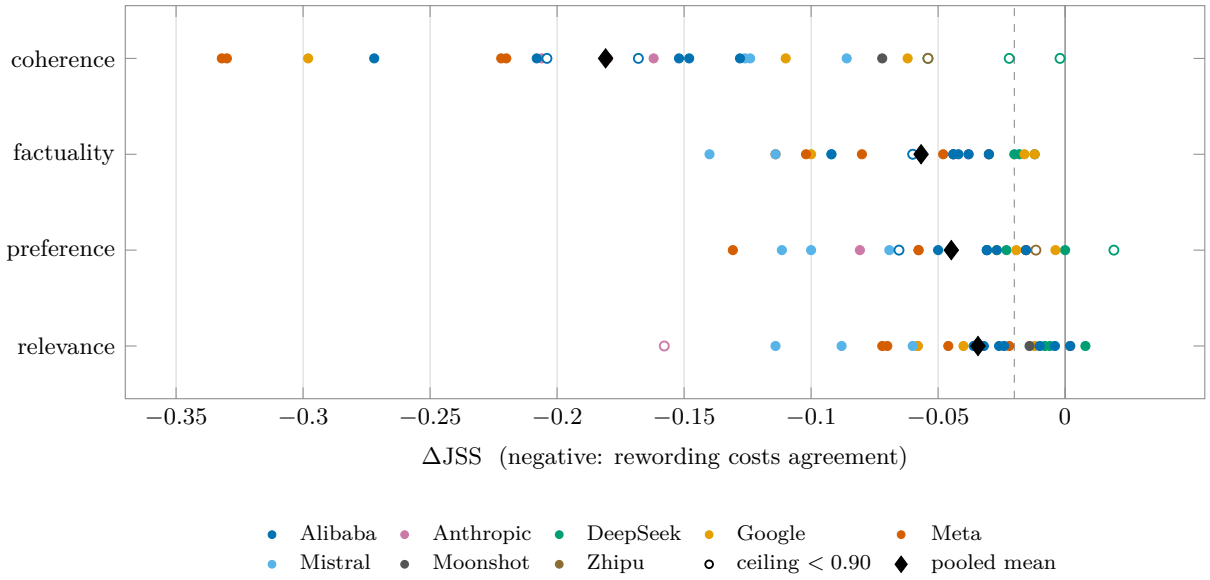


Figure 2: **Paraphrase cost by task across twenty-five judges.** One marker per judge–task cell, coloured by the vendor that trained the checkpoint, which is the unit the confirmatory contrast clusters on. Hollow markers are the eleven cells whose ceiling falls below 0.90. Black diamonds are the pooled mean per task; the solid rule is zero and the dashed rule the −0.02 threshold. Takeaway: the cost is negative almost everywhere, coherence has both the largest loss and much the widest spread, and the three Anthropic judges of the superseded version sit close together, which is what produced its retracted claim that the task matters more than the judge.

5.2 Paraphrasing costs agreement beyond decoding noise

Table 1 reports the endpoint for all ninety-eight cells that carry one (Figure 2); the eleven below the ceiling bar are flagged rather than dropped.

The four confirmatory contrasts pool judges within a task and control the family-wise error rate by Holm–Bonferroni at $\alpha = 0.05$ (Table 2); the individual cells are exploratory under Benjamini–Hochberg, and no confirmatory claim rests on a single cell.

Table 2: **Confirmatory contrasts, judges pooled within a task.** Each row pools the readable cells of one task; p tests a null of no paraphrase cost and is Holm–Bonferroni corrected over the four contrasts. Takeaway: rewording costs agreement on every task, not only on the ordinal one.

Task	Judges	*effect*	Mean ΔJSS	p	Holm
coherence	19		−0.181	1.4×10^{-21}	reject
factuality	24		−0.057	1.6×10^{-13}	reject
preference	20		−0.045	5.3×10^{-8}	reject
relevance	24		−0.034	7.9×10^{-8}	reject

Four rejections establish less than the tally suggests. Under the strict policy, ΔJSS estimates $-\frac{1}{2}\,\mathbb{E}_{\text{item}}\big[\sum_k (p_k - q_k)^2\big]$, where p and q are the judge's response distributions under the two templates. That quantity is non-positive whenever the two prompts induce different distributions, so a point null of exactly zero is false before any data are collected. Rejecting it is a statement about power, and the sign test establishes direction only. The magnitude carries the argument, and we assess it

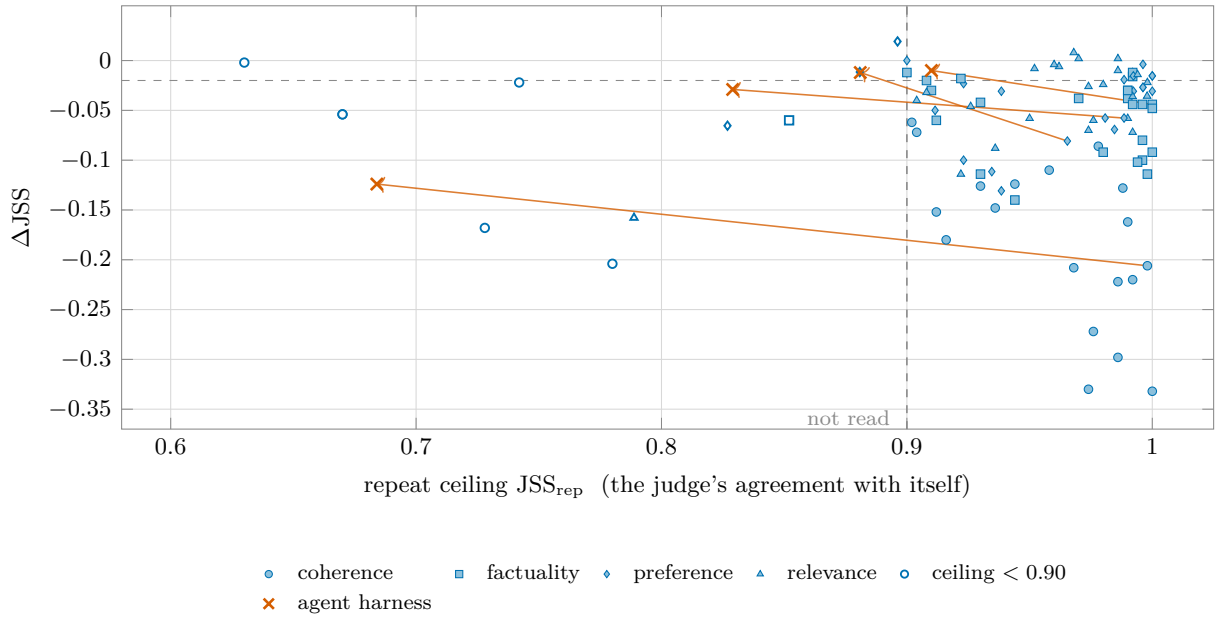


Figure 3: **Repeat ceiling against paraphrase cost, with the agent-harness cells overlaid.** Marker shape gives the task; hollow markers fall left of the 0.90 readability bar. Crosses are Haiku through the agent harness; each arrow runs *from* that judge and task measured through a direct API call *to* the same cell measured through the harness. Takeaway: a low ceiling pulls ΔJSS toward zero rather than away from it, so uncontrolled decoding hides wording sensitivity instead of exposing it, and the harness arrows move left and up for that reason.

against the smallest effect declared of interest (Section 5.3).

The cell treated separately in Appendix C has a ceiling of 0.789, so that judge disagrees with itself on more than a fifth of byte-identical prompts, and sits below even the discarded pilot that set the bar (Section 4.4). Where the ceiling is 0.990 or above the subtraction is nearly a no-op and $\Delta\text{JSS} \approx \text{JSS} - 1$. We collect the repeat arm everywhere regardless, because a judge run at a production temperature above zero has a lower ceiling throughout.

Table 3: **Pooled contrasts with judges clustered by vendor.** This is the confirmatory test. The estimator is the unweighted mean of judge-level ΔJSS, with a cluster-robust (CR1) standard error at the vendor level and t inference on $G - 1$ degrees of freedom; p_{SESOI} is one-sided against the 0.02 threshold, Holm-corrected over four contrasts. Takeaway: treating sibling checkpoints as independent overstates precision by 1.4 to 1.6 times, and two of the four tasks lose their claim to a practically meaningful effect once they are not.

Task	Judges	Vendors	*effect*	Mean	SE (clustered)	p_{SESOI}	Verdict
coherence	19	6		**−0.181**	0.026	0.0008	clears threshold
factuality	24	8		**−0.057**	0.012	0.0076	clears threshold
preference	20	7		−0.045	0.012	0.0416	not established
relevance	24	8		−0.034	0.010	0.1055	not established

The point-null test still rejects on all four tasks under the clustered standard error (p from 9.5×10^{-4} to 1.3×10^{-2}). The unclustered version treats `qwen3-8b`, `qwen3-14b` and `qwen3-32b` as

three independent pieces of evidence about prompt sensitivity, and they are not.

5.3 Effect magnitudes against the threshold

We declared $|\Delta\text{JSS}| < 0.02$ to be not practically meaningful. That threshold was set after the first judge's results were in hand (Section 4), so it is a declared standard and not a pre-registered one. A reader should know which of our thresholds preceded the data and which did not.

Table 4: **Effect magnitude against the threshold.** "Inside threshold" counts cells whose estimate falls within ±0.02; "underpowered" counts those whose minimum detectable effect at their own support exceeds the effect they report. Takeaway: on three of the four tasks roughly half the cells cannot detect an effect of their own size, which is why the confirmatory claims are pooled and clustered (Table 3) rather than counted cell by cell.

Task	Readable cells	*effect*	Pooled ΔJSS	Inside threshold	Underpowered
coherence	19		−0.181	0	0
factuality	24		−0.057	4	10
preference	20		−0.045	6	13
relevance	24		−0.034	9	13

The earlier version of this work, which ran three judges from one vendor, reported factuality as a null on this reading. Twenty-four judges do not support that: only four of its cells fall inside the threshold (Table 4). Relevance is now the task closest to it, with nine of twenty-four cells inside, and we do not report relevance as a null either. Thirteen of its cells cannot detect an effect of their own size at their own support, and a small estimate from an underpowered cell is not evidence of a small effect. Relevance is where the evidence for a practically meaningful cost is weakest; coherence is where it is strongest, with no readable cell inside the threshold and none underpowered.

5.4 Most of coherence's excess is the width of its scale

A judge choosing among five answers has more ways to disagree with itself than one choosing between two, so the comparison against the binary tasks is confounded with the width of the decision space. A reviewer of the first version asked for this control, and it substantially supports the objection. We collapse each rating to *high* (4 or 5) against *low* (1, 2 or 3), on both arms and on the repeat, and recompute the endpoint over the nineteen readable coherence cells.

Table 5: **Coherence with its scale collapsed to binary.** Ratings of 4 or 5 become *high* and 1 to 3 become *low*, on both arms and the repeat. Takeaway: nearly three-fifths of coherence's excess is the width of its decision space rather than the task, though it remains the largest effect of the four.

Coherence, 19 judges	*effect*	Mean ΔJSS
five-point scale		−0.181
binarised		−0.074
shrinkage of the pooled means		59.0%
mean of the per-judge shrinkages		58.6%

We therefore state the headline claim in two parts wherever it appears. As the tasks are actually posed, coherence loses roughly three times what factuality loses. Held at a common decision width,

it loses about a third more. The first number is what a practitioner scoring on a five-point scale should expect; the second isolates the task from its label set. The first is the operational quantity and the second the scientific one; reporting only the first would overstate what is special about judging coherence.

5.5 Sensitivity is larger on the ordinal task than on the others

The two exceptions to the fourteen-of-sixteen tendency are `magistral-small`, which loses 0.140 on factuality against 0.124 on coherence, and `mistral-medium`, which loses 0.092 against 0.086. Both margins are inside the width of their own intervals, so they are better read as ties than as reversals. The ordering at the bottom has also changed: factuality was the smallest effect in the earlier three-judge version, and across twenty-five judges relevance is smaller, so that ordering is superseded rather than refined.

Table 6: **Dispersion across judges against dispersion across tasks.** Within-judge spreads are over the sixteen judges whose four cells are all readable. Takeaway: the two are comparable, which is why the earlier three-judge claim that task dominates judge is withdrawn.

Within task, across judges	spread	Within judge, across tasks	spread
relevance	0.122	minimum (`mistral-small`)	0.026
factuality	0.128	median	0.146
preference	0.131	maximum (`llama3-8b`)	**0.274**
coherence	**0.270**		

That version also concluded that the task matters more than which model does the scoring. At three judges from one vendor the within-task spreads were 0.014, 0.044 and 0.122, against within-judge spreads of 0.124 to 0.162. That comparison measured the similarity of three models from one family. Across six providers the two quantities are comparable, and only two of the sixteen judges have a within-judge spread exceeding the largest within-task spread. On coherence the between-judge spread is the largest dispersion anywhere in this study. We withdraw the claim instead of restating it in a weaker form.

Instability on coherence is milder than a flip between distant ratings. Across the twenty-five coherence cells the quadratic-weighted κ has a median of 0.820, against exact-match agreement from 0.560 to 0.892 with a median of 0.760. The gap is the share of disagreement that is a one-point move, and it is the larger share: `gemma-4-31b` agrees exactly on 0.848 of items but scores 0.941 once distance is weighted.

5.6 Scale does not predict stability within a family

Comparing judges across vendors confounds scale with training recipe, so a reviewer of the first version asked for a same-family ladder. The `qwen-3` family supplies one: three sizes, one provider, one generation, run under the same matched budget.

The ladder is not monotone in any column, and two cautions apply. It is incomplete in an informative direction: the two smaller models have more cells below the readability bar than the largest does, so the smaller a judge is, the less often its own ceiling permits reading its sensitivity at all. A second intended ladder, the `gpt-oss` 20B/120B pair, is absent because its provider's free tier could not finish a cell inside the time we had; we name it instead of leaving the gap unexplained.

Table 7: **The `qwen-3` size ladder.** Three sizes, one provider, one generation, one matched budget; dashes are cells whose repeat ceiling falls below 0.90. Takeaway: a fourfold increase in parameters buys no consistent gain, and relevance reverses at the largest size.

Judge	factuality	coherence	relevance	preference
`qwen3-8b`	−0.060	—	−0.032	—
`qwen3-14b`	−0.042	—	−0.004	−0.050
`qwen3-32b`	−0.044	−0.272	−0.036	−0.027

The results establish something narrower than a scaling law: within one family, parameter count did not predict stability in our suite.

5.7 The tasks discriminate, and coherence only narrowly

Section B.3 established that no heuristic beats chance. The complementary check requires the best judge on each task to exceed a threshold, measured against the majority-class rate, not against uniform chance.

Table 8: **Discrimination against the majority-class rate.** A constant-answer judge already attains the majority-class rate, so quoting uniform chance would overstate every margin. Takeaway: three tasks discriminate comfortably and coherence does not, and coherence carries the largest paraphrase effect.

Task	threshold	best observed	majority class	judges clearing
factuality	0.75	0.960 (`gemini-3.7-flash`)	0.500	21/25
relevance	0.70	0.899 (`deepseek-v4-pro`)	0.500	25/25
preference	0.65	0.892 (`gemma-4-31b`)	0.500	24/25
coherence	0.40	0.452 (`claude-sonnet`)	0.348	6/25

Two reviewers of the first version held that three of the four tasks failed to discriminate. At twenty-five judges the opposite holds, and the task that reverses is not the one they expected. Coherence's best judge exceeds a constant-"4" judge by ten points, and at $n = 250$ the interval on 0.452 reaches below the 0.40 threshold, so coherence is only weakly discriminating. That and its large paraphrase effect follow from the same property: a five-point scale over gold labels that are a rounded three-annotator mean.

Answer-A rates confirm that the swap design holds against live judges. Across the fifty pairwise cells the rate spans 0.045 to 0.600, with a median of 0.518 on relevance and 0.504 on preference. A reviewer of the first version reported that almost every judge returned label A on every pairwise item, which saturated the metric; under the swap no cell saturates at the high end. The one degenerate cell is `claude-sonnet` on preference at 0.045, and it is parse failure, not position behaviour: it carries a 43% malformed rate on the support it has, and it fell below the support floor. We leave it in the range, because excluding the one cell that embarrasses the design would be the wrong precedent.

5.8 The same judge, run inside an agent harness

Haiku was re-run on all four tasks through an agent harness that issues the same items in batches of fifty and does not expose temperature.

Table 9: **The same judge through a direct API call and through an agent harness.** Model, items, templates and repeat design are identical; only the transport differs. Takeaway: both terms fall under the harness but the ceiling falls faster, so the measured sensitivity shrinks rather than grows.

Task	Transport	JSS	JSS_{rep}	*effect*	ΔJSS
factuality	direct API	0.956	1.000		−0.044
	agent harness	0.900	0.910		−0.010
coherence	direct API	0.792	0.998		−0.206
	agent harness	0.560	0.684		−0.124
relevance	direct API	0.930	0.990		−0.058
	agent harness	0.800	0.829		−0.029
preference	direct API	0.896	0.965		−0.081
	agent harness	0.869	0.881		−0.012

We report the harness as a second condition, not as a judge. Its cells are excluded from Table 1 and from every claim above, because they carry no declared decoding configuration and are comparable only to themselves across arms. Three alternative explanations were ruled out. All 102 batches returned their full complement of labelled items, no answer failed to parse, and the repeat prompts were verified byte-identical to the arms they baseline. One batch did come back eight items short; it was rejected and re-issued, since a silently shortened batch would shift every later label within it.

The scope of the contrast is one model on four tasks. Only Haiku is reachable under both transports, because the harness resolves its Sonnet and Opus aliases to a later generation than the ones evaluated here. Two further judges were run through the same harness on coherence and factuality, the tasks where the collapse is largest and smallest. Neither has an API counterpart, so neither supports a paired contrast.

Table 10: **Three judges through the same harness, on the task where the collapse is largest and the task where it is smallest.** Neither of the two additional judges has an API counterpart, so neither supports a paired contrast. Takeaway: the ceiling collapse is task-dependent, not model-dependent, and it is not ordered by model capability.

Task	Judge (agent harness)	JSS	JSS_{rep}	*effect*	ΔJSS
factuality	Haiku 4.5	0.900	0.910		−0.010
	Sonnet (later)	0.980	0.970		+0.010
	Opus (later)	0.980	0.990		−0.010
coherence	Haiku 4.5	0.560	0.684		−0.124
	Sonnet (later)	0.520	0.616		−0.096
	Opus (later)	0.696	0.848		−0.152

A harness that simply injected decoding noise would have degraded both tasks. Instead it appears to amplify borderline-ness the task already carries. This is the reading two tasks and three models will bear, and not more: it does not establish a mechanism, and three models from one

vendor on two tasks cannot settle the question for harnesses in general.

5.9 Limits of this evidence

Twenty-five judges across six providers is the breadth this design was built for and could not report in its first version, but the roster is weighted towards small and mid-sized checkpoints: three flagship models were dropped on cost and four more were withdrawn when their providers' free tiers proved too slow to finish a cell. More consequentially, *it contains no OpenAI model*. The GPT family was to be reached through a subscription harness, not the paid API, and that route carries no declared decoding configuration, the same objection that keeps our own agent-harness cells out of Table 1 (Section 5.8). Mixing one vendor measured a different way into a contrast that clusters by vendor would have been worse than leaving it out. Every cross-vendor statement here is about the eight vendors in Table 3, and the scale conclusions rest on two size ladders, not on a sweep of the largest models available.

Two of the four tasks do not clear the threshold once judges are clustered by vendor (Section 5.3): preference falls just outside after correction and relevance is not close, so on those two the practical claim is unestablished, not merely small. Nor is the construct assumption independently validated. No judge, human or model, scored the forty instruction pairs for this release; the paper offers a deterministic structural audit and the forty pairs in full (Section 3).

Intervals cluster at the item, the right unit for the arms and repeats nested inside an item, but items are themselves nested in source records, on every task (Section 3.1). The 250 relevance items come from 50 TREC-COVID topics, the 250 coherence items from 92 SummEval documents, the 250 factuality items from 125 TruthfulQA records and the 130 preference items from 68 MT-Bench questions. Every task therefore has an effective sample smaller than its item count suggests, relevance most of all; Appendix C.2 recomputes every pooled interval at the record and no conclusion changes. The five templates are a fixed set, not a sample, so every interval conditions on these particular strings and the results describe sensitivity to *these* rewordings.

Five cells are measured on terms the rest are not. The preference split is complete for twenty-three of the twenty-five judges. Sonnet and Opus stopped at 44 and 170 of 260 rows when the API balance was exhausted mid-run, and the pipeline refuses to emit ΔJSS below the declared floor of 100 clusters, so those two cells report no endpoint rather than one computed from 17 and 80 clusters. Separately, Opus 4.7 rejects an explicit temperature parameter and ran at the provider default while the other twenty-four judges ran at zero, so its three cells that do carry an endpoint are not measured under the decoding policy the rest are; this is recorded on every one of its call records, never inferred.

Every judge but that one is run at temperature zero, where the repeat ceiling is highest. Production judges are often run above zero, or inside a harness that fixes decoding on the caller's behalf. We expected the endpoint's separation to grow under that noise; the one uncontrolled-decoding condition we have measured (Section 5.8) does the reverse, the ceiling falling faster than paraphrase agreement so that ΔJSS contracts and a real effect becomes harder to detect.

Every prompt and every source corpus is English, so whether a judge stable in English stays stable under translation-equivalent prompts is untested, and it is a likely place for rankings to change. Every judge is also reached through a single provider: alias drift, replica-level routing or an undocumented checkpoint update would move every number in this paper at once, and we could not detect it against an independent host. We record the provider-reported model string on every call so such a change is visible after the fact.

6 Conclusion

We introduced JudgeSense, a benchmark of 880 items asking whether a judge returns the same verdict when the same request is worded differently, scored against that judge's own agreement with itself. Rewording costs agreement on all four tasks across twenty-five judges. On two of them the cost clears the threshold we declare for a practically meaningful effect, once sibling checkpoints stop counting as separate evidence, and the ordinal task loses 0.181 where the next loses 0.057. We recommend that judge stability be reported against a repeat baseline, measured in the transport the judge will run in, and never read from a cell whose ceiling is low.

Acknowledgements

Acknowledgements are omitted from the review copy and will be added for the camera-ready.

Use of AI assistance

Every quantitative claim in the paper is regenerated from committed artifacts by a released script rather than transcribed, and the authors have verified them against those artifacts. The authors take full responsibility for the content, the analysis, and any remaining errors.

Data and code availability

Everything needed to recompute the paper is released.

- **Benchmark.** The 880 items across the four tasks, with both instruction templates per item, the position-swapped orderings for the two pairwise tasks, and the twenty templates in full.
- **Raw judge outputs.** Every call record behind every number: 31,194 rows over 100 judge–task cells, each carrying the resolved decoding parameters and the provider-reported model string.
- **Analysis.** The metric, the item- and source-clustered bootstraps, the multiplicity plan with its commit history, and the scripts that emit every table and both data figures in this paper.
- **A standalone scorer** that computes the endpoint for a judge that is not in our roster, from a predictions file, with no API keys and no dependency on our provider clients.
- **The heuristic baselines** that would defeat each task were the construction flawed, together with the scores they actually reach.

Code: `https://github.com/rohithreddybc/judgeSense`. Dataset: `https://huggingface.co/datasets/Rohithreddybc/judgesense-benchmark`.

Reproducibility statement

The artifacts listed above are released in full. Every table and both data figures in Section 5 are regenerated from the committed raw outputs by a single script. Descriptive counts about the construction, such as how many source comparisons each filter removed, are reported from the loader's own logged output. Each split is content-hashed, and a 32-check audit gate runs in continuous integration. Judge model identifiers, resolved decoding parameters and the provider-reported model string are recorded on every call record, so configuration drift is detectable after the fact rather than assumed absent.

A The polarity defect, and why no remapped analysis is reported

The polarity constraint in Section 3 corrects a defect of our own earlier design, which included a factuality template that inverted the answer polarity ("does this response contain factual errors? answer NO if accurate"). Under a single shared label space "YES" then denoted opposite conclusions in different arms, so a judge answering both arms correctly was recorded as disagreeing with itself. That is a label-convention defect in the benchmark, not a property of any model, and it measures a different construct: handling a polarity flip is a comprehension task, whereas we ask whether equivalent phrasings yield equivalent verdicts. Excluding such pairings biases the figures in a knowable direction, since polarity-inverted pairings are among the hardest items in the set and removing them can only push measured consistency upward.

The alternative is to retain inverted templates and remap each template's raw answers onto one canonical decision space before comparison. That machinery is implemented and released, but no remapped analysis is possible here, and not because we are deferring the work: the v2 construction removes polarity inversion at source, so all twenty shipped templates request the same answer space in the same direction and there is nothing left for the remap to operate on. Building a polarity-inclusive split would be a new release, and the machinery is available for anyone who wants one.

The earlier version did quantify what the exclusion costs. Its magnitude rested on data we have withdrawn, but the shape of the result stands. Including polarity-inverted pairings produced an apparent flip rate near-identical across every judge tested, and a rate that does not vary with the model is a property of the instrument, not of the models. That uniformity is what identified the template as a label-convention defect rather than a finding about judges. We keep the inference and discard the number.

B Discussion, construction detail and impact

The body carries the measurement. This appendix carries what supports it: what the results mean for a team choosing a judge, the shortcut and outcome-category controls behind the construction, and the statement on broader impact.

B.1 Discussion

B.1.1 What the result means for practice

On the ordinal task, choosing a stronger judge does not reliably buy stability, but neither does it fail to; the two moves are comparable in size. Across the nineteen readable coherence cells the spread between the most and least sensitive judge is 0.270, the largest dispersion in this study and wider than the median judge's spread across all four tasks (0.146; Table 6). A practitioner choosing a coherence judge is making a decision that matters at least as much as the decision about response format, which is the opposite of what the earlier version of this work concluded from three judges of one vendor (Section 5.5).

The obvious mechanism is the size of the answer space: a five-point scale gives a rewording more ways to move the answer, and the weighted κ shows most of that movement is a single scale point. Collapsing the scale at ≥ 4 removes 58.6% of the coherence effect, averaged over judges (Section 5.4). What survives still leads the four, so the ordering holds, but the margin is about 1.3 times factuality, not three.

The same confound cuts the other way. A binary answer space quantises the judge's latent uncertainty: a move from "YES" to a less confident "YES" is recorded as perfectly consistent, while the same drift on a five-point scale crosses a category boundary and is recorded as a flip. Relevance, the smallest effect in the study at -0.034 and the task with the most cells inside the declared threshold, is therefore consistent with two readings this design does not separate: either judges really are stable when picking between two passages, or a two-valued answer space cannot show the instability that is there. A small binary effect is not licence to trust binary judges more.

B.1.2 Refusal is an evaluation failure mode, not an aside

One judge declined a quarter of the relevance items while another, on identical prompts, declined none. For anyone using an LLM judge over medical or otherwise sensitive corpora this is a first-order problem: the missing verdicts are not missing at random, and a harness that silently treats a decline as a parse failure will report a format-adherence problem that does not exist. A rewording can also change whether a judge answers at all, which makes refusal an outcome of the manipulation, so conditioning on the answered subset is selection on a post-treatment variable. Worst-case bounds accompany the conditioned estimate for that reason, and where refusals are common those bounds are wide.

B.1.3 On building a benchmark that can be wrong

The defects in our own rebuild are the most transferable part of this work. Each passed every check that existed at the time. A deterministic rotation assigning template pairs to items ran in lockstep with the alternating emission of correct and incorrect answers, which made the template pair a perfect answer key and skewed two templates' label balance to 75/25, so a constant-answer judge would have appeared to hold a large template preference. Candidate responses reused across pairings carried contradictory gold labels, so a judge reasoning correctly and consistently was scored wrong on nearly half the preference items. A length-balancing step that correctly removed a verbosity shortcut cut only one of the two buckets, and cut it by vote margin, leaving the buckets matched on length and far apart on annotator agreement.

None of these is visible in labels, provenance, or aggregate statistics. All three were found by asking what a heuristic with no understanding would score, and by checking whether the artifact contained a rule that predicted the answer. We would encourage that as routine: before reporting what models score on a benchmark, report what nothing scores on it, with an interval and a support count.

B.1.4 Limitations

Section 5.9 states what this evidence does not cover. Two points bear repeating where the practical reading is made. The task ordering is a strong tendency, not a rule: it holds for fourteen of the sixteen judges whose four cells are all readable, and the two exceptions lose slightly more on factuality than on coherence. And the effects are modest on three of the four tasks. Only coherence and factuality clear the 0.02 threshold once judges are clustered by vendor; nine of twenty-four relevance cells and six of twenty preference cells sit inside it, against four of twenty-four on factuality, and thirty-six of the eighty-seven readable cells cannot detect an effect of their own size. The ordinal result survives unambiguously. Coherence loses roughly three times what the next task loses on its own scale, and about a third more than factuality once that scale is collapsed.

B.2 Broader impact statement

The work is a measurement instrument for evaluation infrastructure, and its risks are mostly the risks of being believed too readily.

A benchmark that reports stability invites the reading that a stable judge is a good one. It is not: JSS never consults the ground-truth label, and a judge that is wrong on every item but wrong identically scores perfectly. Selecting a judge on stability alone would optimise for a model that is confidently and reproducibly incorrect. Stability is necessary for an instrument and nowhere near sufficient.

The refusal result has a distributional edge. One judge declined a quarter of the relevance items, drawn from a medical corpus, while another on identical prompts declined none. Where an evaluation harness silently drops declines, the items lost are not lost at random, and the topics most likely to be dropped are the sensitive ones. A pipeline that scores refusals as format failures will report a clean evaluation over a corpus it partly refused to read. That failure is quiet, and it gets quieter as the corpus gets more sensitive.

The benchmark is built from four existing English corpora and inherits whatever they carry. Coherence gold is a rounded three-annotator mean, preference gold is a human majority vote, and both encode the judgements of particular annotator pools. Nothing here supports a claim about judge behaviour outside English, and we have not measured it.

We see no dual-use concern specific to this artifact. It measures agreement between two paraphrases of the same request; it does not generate content, elicit capabilities, or make a model easier to misuse.

B.3 Shortcut controls

We evaluate the shipped splits against judges that use no judgment at all, run over the released files, not the loaders, so these are properties of the artifact. Position only (always answer A) scores 0.500 on relevance ($n = 500$) and 0.500 on preference ($n = 260$), by construction of the swap design. Length only (pick the longer candidate) scores 0.482 on relevance ($n = 494$) and 0.508 on preference ($n = 256$). Lexical overlap only (pick the candidate sharing more terms with the query) scores 0.540 on relevance ($n = 252$). Support counts fall below the row counts because tied rows are excluded rather than split. All three come from `scripts/shortcut_controls.py`, whose output is committed.

None exceeds chance in either direction. The overlap control is the weakest: its 95% Wilson interval is $[0.478, 0.600]$, so the data are consistent with a residual lexical signal worth about 0.57, and it reads only half the relevance rows because the remainder tie on the overlap statistic.

The length control is an enforced property of the item set. Annotators in the preference corpus preferred the longer response in roughly 70% of usable comparisons, so we hold the winner-longer share at exactly 50% by taking an equal number of items from each length bucket. That 50% is a construction invariant. The 0.508 above is what a length-following judge scores on the rendered rows once ties are dropped.

The preference split ships 130 items, not the round 250: of the source comparisons, 607 failed the decisive-vote label rule or carried no strict majority, one was duplicate content, 67 had gold contradicting another item's gold on shared candidate text, and 102 more went to hold the length balance exactly. The last two filters were added after we found both failures in our own shipped data.

Balancing on length alone would install a second confound, because only the winner-longer bucket is cut and so any cutting criterion applies to one bucket and not the other. Cutting by vote margin,

as an earlier build did, left the buckets matched on length but far apart on annotator agreement (mean vote-margin ratio 0.567 against 0.745, 16% against 54% unanimous), so a judge stronger on close calls would read as length-biased. We therefore *stratify*: the winner-shorter bucket fixes the target distribution of vote shapes, absolute margin and total votes, and the winner-longer bucket is filled stratum by stratum to match it. The shipped buckets agree to 0.0003 in mean vote-margin ratio (0.8603 against 0.8605), have identical medians, and contain 49 unanimous comparisons each.

The opposite failure is equally real. The filter that removes contradictory gold is a greedy selection over a conflict graph, so the order in which items are considered determines which survive and how hard they are. Keeping the firmest human majorities first is worse on both counts in this pool, giving 106 items at 90.6% unanimous against 130 items at 75.4% when the most contested are kept first, and a split on which annotators were nearly always unanimous cannot separate a competent judge from a mediocre one. We resolve conflicts contested-first.

The lexical control required a similar correction. Selecting each distractor to maximise its term overlap with the topic made distractors *denser* in query terms than the relevant passages, so "pick the passage with *lower* overlap" recovered the answer far above chance. Matching the distractor's retrieval score to the positive's removes overlap as a signal in either direction, which is why the control is reported two-sided.

B.4 Outcome categories: verdict, refusal, malformed

Each call terminates in exactly one of three outcomes. A **verdict** is a response the strict parser maps to a label in the task's answer space. A **refusal** is a response the provider itself flags as declined, typically with empty content; providers spell this differently (`stop_reason="refusal"` for Anthropic, `finish_reason="content_filter"` for OpenAI-compatible APIs, `SAFETY` or `RECITATION` for Google), so the accepted set is enumerated in one place in the released code, matched case-insensitively, with the raw string retained on every record. Recognising only one family's spelling would report a refusal rate of zero for every other family and silently charge their declines as paraphrase disagreement. **Malformed output** is a completed response the parser cannot map, recorded with its raw text and never rounded to the nearest label.

The two non-verdict outcomes are treated differently. A malformed arm is charged as disagreement and stays in the support, because the model completed a response and the pair reflects a judgment the parser could not read. A refused arm is removed from the support, because the provider halted before any judgment was rendered: scoring a refusal as paraphrase disagreement asserts that the judge produced two conflicting judgments, which it did not, and treating refusal as a third label is worse still, since a judge that refuses everything would then score JSS = 1.0.

JSS is computed over pairs in which both arms returned verdicts and report the support size n alongside every figure. Refusal is reported per cell as three quantities: the refusal rate (refused calls over all calls), the refusal discordance rate (RDR; pairs in which exactly one arm was refused), and the consistent refusal rate (pairs in which both arms were refused). RDR is itself a sensitivity statistic. A nonzero RDR means a meaning-preserving rewording changed whether the judge was willing to judge at all, which is invisible to any analysis that scores refusals as disagreement or discards them.

Conditioning on verdict pairs restricts a refusing judge to the items it agreed to judge, so judges are compared on different subsets. We answer that with a measurement: every cross-judge comparison in a task with nonzero refusals is recomputed on the common support, the items every compared judge answered under both phrasings, and beside it we report a refusal-inclusive figure in which every refused arm counts as disagreement, the most punitive reading.

C Supporting analyses

These are the robustness checks and single-cell diagnostics behind the findings in Section 5.1. They are reported in full here instead of in the body, since a reader does not need them to follow the argument, and because several answer requests made of the previous version of this work.

C.1 Testing against the threshold rather than against zero

The contrasts above reject a null of no effect, and that null is weaker than it looks. Under the strict policy ΔJSS estimates a quantity that is non-positive for any two prompts that do not induce identical response distributions, so a point null of exactly zero is false before any data are collected and rejecting it reports statistical power, not a property of the judge. The test that carries the claim asks whether the effect is larger than the smallest difference we declared of interest, and we report it in the form a reader should hold us to: one-sided, $H_0 : \overline{\Delta\text{JSS}} \geq -0.02$, Holm-corrected over the same four contrasts (Table 11).

Table 11: **Testing against the threshold rather than against zero, treating judges as independent.** One-sided, $H_0 : \overline{\Delta\text{JSS}} \geq -0.02$, Holm-corrected over four contrasts. Table 3 is the same test with judges clustered by vendor, and is the confirmatory one. Takeaway: all four tasks clear the threshold here and only two survive clustering, so the independence assumption is doing most of the work.

Task	Judges	*effect*	Mean	Margin beyond 0.02	z	p (Holm)
coherence	19		−0.181	−0.161	−8.47	1.2×10^{-17}
factuality	24		−0.057	−0.037	−4.77	9.4×10^{-7}
preference	20		−0.045	−0.025	−3.02	1.3×10^{-3}
relevance	24		−0.034	−0.014	−2.23	1.3×10^{-2}

All four survive this version of the test, but the test assumes each judge is an independent draw, and Section 4.3 explains why they are not. Under the vendor-clustered standard error we report as confirmatory (Table 3), only coherence and factuality clear the threshold. Preference falls just outside after correction and relevance is not close. The ordering is the same either way, and the ordering is the point: coherence clears by a wide margin under both, relevance clears under neither once sibling checkpoints stop being counted separately. Had we set the threshold at 0.05 instead of 0.02, only coherence would clear it.

C.2 The conclusions survive clustering at the source record

Intervals elsewhere in this paper cluster at the item, the right unit for the two arms and two repeats nested inside one item. It is not the only nesting present, and all four tasks carry a second one. The 250 relevance items are drawn from 50 TREC-COVID topics and the 250 coherence items from 92 SummEval documents; the 250 factuality items come from 125 TruthfulQA records, each supplying one accurate and one inaccurate statement, and the 130 preference items from 68 MT-Bench questions. Two items sharing a record are not independent draws, so item-clustered intervals are too narrow wherever that dependence is strong. We reported this as a limitation of the first version, and it is recomputable from the committed rows without issuing a single call.

Resampling source records instead of items, 2,000 draws, gives Table 12.

Every interval excludes zero, and every interval excludes the 0.02 threshold as well. Relevance is where the nesting is heaviest, five items per topic on average; factuality is where it is lightest, at

Table 12: **Intervals clustered at the source record rather than the item.** Resampling records with replacement, 2,000 draws. Takeaway: the nesting of items inside documents and topics changes no conclusion, so the item-clustered intervals reported elsewhere are not doing the work through an understated denominator.

Task	Records	Unit	*effect*	Mean	95% CI (source-clustered)
coherence	92	SummEval documents		−0.179	[−0.191, −0.167]
factuality	125	TruthfulQA questions		−0.058	[−0.067, −0.049]
preference	68	MT-Bench questions		−0.052	[−0.066, −0.040]
relevance	50	TREC-COVID topics		−0.038	[−0.046, −0.030]

two items per record. No pooled mean moves by more than 0.001 and no interval widens by more than 0.002.

An earlier version of this analysis read the clustering unit off the item naming scheme and reported factuality and preference as un-nested, at 250 and 130 records. Both are nested, the dataset says so in a provenance field the script did not consult, and for those two tasks what was printed as a source-clustered interval was the item-clustered interval under another name. The table above is computed from `source.source_record_id`, and the script now fails rather than falling back to the item when a record identifier does not parse.

We report this because the alternative is asking a reader to accept that a known dependence does not matter. It does not matter here, and we measured that instead of assuming it.

C.3 No judge wins by collapsing onto one label

A reviewer of the first version observed that our endpoint rewards compressed output: a judge that answers "YES" to everything agrees with itself perfectly under any rewording, and would score a perfect JSS for the worst possible reason. The repeat ceiling does not catch this, because such a judge is genuinely reproducible. The discrimination check in Section 5.7 catches it only at the extreme. We therefore record the decision distribution of every cell and report it here.

Table 13 reports decision entropy, in bits, over the outcomes each cell actually produced.

Table 13: **Decision entropy per task, in bits.** Computed over the outcomes each cell actually produced, across twenty-five judges. The last column is the entropy of a uniform distribution over the task's decision space, which an observed value can exceed where a cell also produced refusals or malformed output. Takeaway: no judge wins by collapsing onto a single label, so the endpoint is not rewarding compressed output.

Task	minimum	median	maximum	maximum possible
factuality	0.944	1.000	1.394	1.000
coherence	1.031	1.751	2.076	2.322
relevance	0.981	1.052	1.541	1.000
preference	0.881	1.018	1.393	1.000

No cell is close to collapse. The lowest entropy anywhere is 0.881 bits (`magistral-small` on preference), against 1.0 for a judge splitting a binary decision evenly; a judge answering one label on every item would score 0. On coherence, thirteen of the twenty-five judges use all five scale points and the remaining twelve use four, so no judge treats the ordinal task as binary.

Two of the maxima exceed the binary ceiling. Entropy is computed over the outcomes a cell produced, and some cells produced an outcome outside the declared label set: `claude-sonnet` returned `UNCLEAR` on four factuality arms, and five judges emit it on coherence alongside the five ratings. Those are counted as malformed in Section C.4 and never as a verdict, so the endpoint is unaffected; the entropy column simply shows them.

C.4 Refusal and malformed output are separate failures, and Sonnet has both

Sonnet declined 24.7% of relevance arm calls, reported by the provider as a refusal with empty content. Opus refused 5.1% of the same prompts and Haiku none at all. The corpus is TREC-COVID, so the items are medical, and the effect is a property of one judge's safety behaviour, not of the benchmark: the three judges saw identical prompts.

The outcome categories partition that cell's 1,000 arm calls into 583 verdicts, 249 refusals and 168 malformed responses. A harness that folded refusals into malformed output would report a 41.7% format-adherence failure. The true malformed rate, computed over the arms the judge actually attempted, is 22.4%. That is still a substantial and genuine instruction-following failure, and the refusal finding should not be allowed to obscure it.

Conditioning on the answered subset raises the fair objection that the three judges are then scored on different items. We answer it by recomputing on the common support, the 187 of 250 relevance items every judge answered under both phrasings. The point estimates barely move: -0.008 for Haiku, -0.001 for Opus and $+0.003$ for Sonnet. The conditioning is not what produces the difference between these judges.

The deeper problem is not shared support. Because a rewording can itself change whether a judge answers, refusal is one of the manipulation's outcomes and not a nuisance to be dropped, and conditioning on the answered subset is selection on a post-treatment variable. We report worst-case bounds. For this cell the conditioned JSS is identified only within $[0.486, 0.746]$, a width of 0.26 over 130 unidentified pairs. That region is wide, and we would still rather show it than quote a point estimate resting on an assumption nobody can check.

C.5 Why Sonnet's relevance cell is not evidence about wording

Three facts about that cell arrive together. Its repeat ceiling is 0.789, the lowest we observe. Its malformed rate differs by a factor of 2.7 between the two arms (0.327 against 0.122). And its arm-specific repeat agreement differs by 0.209 (0.685 under one template, 0.893 under the other).

That last figure is the diagnostic we built to detect templates that are not exchangeable, and here it fires. If one arm fails to parse three times as often as the other, the disagreement between arms is substantially a difference in parse surface, not a difference in judgement, and $\Delta\text{JSS} = -0.158$ cannot be read as sensitivity to wording. The paraphrase audit anticipated this: it warns that two of the relevance templates state no single-token answer constraint, so their parse surfaces differ from their partners'.

This cell is excluded from the ordering and spread claims above, and report it as what it is: a judge whose behaviour on this corpus is dominated by refusal and parse failure. It stays in Table 1 and in the count of intervals excluding zero, flagged rather than deleted.

D Experimental setup

Every judge below was reached through the provider named, at `temperature` $= 0$ where the provider accepts it, with an identical output budget of 1,024 tokens and explicit suppression of reasoning

traces where the provider exposes that control. Whether suppression was requested and whether it was honoured are recorded per call in the released logs, never inferred. One judge rejects an explicit temperature parameter and therefore ran at its provider default; its cells are marked in Table 1.

Table 14: **The twenty-five reported judges.** Alias as used throughout the paper, the vendor that trained the checkpoint, the checkpoint identifier sent to the provider, and the provider that served it. Vendor and provider differ wherever a model is served by a host that did not train it. Takeaway: every alias in Table 1 resolves to a pinned checkpoint string.

Alias	Vendor	Checkpoint	Provider
`claude-haiku`	Anthropic	`claude-haiku-4-5-20251001`	anthropic
`claude-opus-4-7`	Anthropic	`claude-opus-4-7`	anthropic
`claude-sonnet`	Anthropic	`claude-sonnet-4-5`	anthropic
`deepseek-v4-flash`	DeepSeek	`deepseek/deepseek-v4-flash`	novita
`deepseek-v4-flash-ds`	DeepSeek	`deepseek-v4-flash-0731`	dashscope
`deepseek-v4-pro`	DeepSeek	`deepseek-v4-pro-0813`	dashscope
`gemini-3.7-flash`	Google	`gemini-3.7-flash`	google
`gemini-flash`	Google	`gemini-2.5-flash`	google
`gemma-4-31b`	Google	`google/gemma-4-31B-it`	huggingface
`glm-5.2`	Zhipu	`glm-5.2`	dashscope
`kimi-k3`	Moonshot	`kimi-k3`	dashscope
`llama-3.3-70b`	Meta	`meta-llama/Llama-3.3-70B-Instruct`	huggingface
`llama-4-maverick`	Meta	`meta-llama/Llama-4-Maverick-17B-128E-Instruct-FP8`	huggingface
`llama-4-scout`	Meta	`meta-llama/Llama-4-Scout-17B-16E-Instruct`	huggingface
`llama3-8b`	Meta	`meta-llama/Llama-3.1-8B-Instruct`	huggingface
`magistral-small`	Mistral	`magistral-small-latest`	mistral
`mistral-medium`	Mistral	`mistral-medium-2604`	mistral
`mistral-small`	Mistral	`mistral-small-latest`	mistral
`qwen`	Alibaba	`qwen/qwen-2.5-72b-instruct`	novita
`qwen-3.6-flash`	Alibaba	`qwen3.6-35b-a3b`	dashscope
`qwen3-14b`	Alibaba	`Qwen/Qwen3-14B`	huggingface
`qwen3-32b`	Alibaba	`Qwen/Qwen3-32B`	huggingface
`qwen3-8b`	Alibaba	`Qwen/Qwen3-8B`	huggingface
`qwen3.7-flash`	Alibaba	`qwen3.7-flash-2026-07-15`	dashscope
`qwen3.8-27b-hf`	Alibaba	`Qwen/Qwen3.8-27B`	huggingface

Judges registered but not reported. 4 checkpoints are in the registry and carry no cell in Table 1. None was dropped for its answers; each was withdrawn on throughput, because the provider's free tier could not finish a cell in a usable time. They are listed so the roster is auditable rather than curated.

Alias	Vendor	Why it was withdrawn
`gemini-3.1-pro`	Google	250 requests/day, 16.3 days to finish
`gpt-oss-120b`	OpenAI	200k tokens/day/model, 7.0 days to finish
`gpt-oss-20b`	OpenAI	200k tokens/day/model, 5.3 days to finish
`qwen3.8-27b`	Alibaba	200k tokens/day/model, 1.5 days to finish

E The twenty instruction templates

Every template shipped with the benchmark is printed below, exactly as the builder emits it. Placeholders in braces are filled per item; the payload is otherwise unchanged between the two arms of a pair.

Each item is issued under exactly two of the five templates for its task, assigned by a label-stratified seeded permutation over the ten unordered pairs (Section 3). No item appears under more than one pair, so template identity is nested inside item identity and no between-template contrast is identifiable from this design.

The audit described in Section 3 is run against these strings: it checks that the requested answer set is identical within each pair, that no template inverts polarity, that each names its own task's construct, and that no pair is a near-duplicate. It cannot establish that a competent reader maps two wordings onto the same question, so the equivalence claim is stated as an assumption and not as a result.

E.1 Factuality (TruthfulQA)

Requested answer set: {YES, NO} for all five templates.

T1. `Is this statement factually correct? Answer YES or NO only. {text}`

T2. `Fact-check the statement below. Reply YES (correct) or NO (incorrect). {text}`

T3. `Is the following statement accurate? Answer with one word: YES or NO. {text}`

T4. `Evaluate the factual accuracy of this statement. Respond YES if it is correct, NO if it is not. {text}`

T5. `Judge whether the statement is factually correct. Your entire answer must be YES or NO. {text}`

E.2 Coherence (SummEval)

Requested answer set: $\{1, \ldots, 5\}$ for all five templates.

T1. `Rate the coherence of this text from 1 (incoherent) to 5 (very coherent). Reply with one number only. {text}`

T2. `How coherent is the following text? Score it 1-5 (1 = incoherent, 5 = very coherent). Number only. {text}`

T3. `Assign a coherence rating from 1 to 5, where 1 means incoherent and 5 means very coherent. Answer with a single digit. {text}`

T4. `On a 1-5 coherence scale (1 = incoherent, 5 = very coherent), what score does this text deserve? Respond with the number only. {text}`

T5. `Score this text's coherence: 1 (incoherent) through 5 (very coherent). Give only the number. {text}`

E.3 Relevance (BEIR TREC-COVID)

Requested answer set: {A, B} for all five templates.

T1. `Which passage is more relevant to the query "{query}"? Answer A or B only. A: {first} B: {second}`

T2. `For the query "{query}", which passage is more relevant? Reply with A or B. A: {first} B: {second}`

T3. `Query: "{query}". Choose the more relevant passage. Answer A or B. A: {first} B: {second}`

T4. `Considering the query "{query}", decide which passage is more relevant. Respond A or B only. A: {first} B: {second}`

T5. `Given the query "{query}", pick the passage with higher relevance. Your answer must be A or B. A: {first} B: {second}`

E.4 Preference (MT-Bench human judgments)

Requested answer set: {A, B} for all five templates.

T1. `Which response to "{query}" is better? Answer A or B only. A: {first} B: {second}`

T2. `For the question "{query}", which response is of higher quality? Reply with A or B. A: {first} B: {second}`

T3. `Question: "{query}". Choose the better response. Answer A or B. A: {first} B: {second}`

T4. `Considering the question "{query}", decide which response is better. Respond A or B only. A: {first} B: {second}`

T5. `Given the question "{query}", pick the stronger response. Your answer must be A or B. A: {first} B: {second}`

References

[1] Jihyun Janice Ahn and Wenpeng Yin. Prompt-reverse inconsistency: LLM self-inconsistency beyond generative randomness and prompt paraphrasing, 2025. URL https://arxiv.org/abs/2504.01282.

[2] Negar Arabzadeh and Charles L. A. Clarke. A Human-AI comparative analysis of prompt sensitivity in LLM-based relevance judgment. In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2025. doi: 10.1145/3726302.3730159.

[3] Rohith Reddy Bellibatlu, Edward Raff, and Wenbin Zhang. JudgeSense: A benchmark for prompt sensitivity in LLM-as-a-Judge systems, 2026. URL https://arxiv.org/abs/2604.23478. Earlier version of this benchmark; superseded by the present work.

[4] Cheng-Han Chiang and Hung-yi Lee. Can large language models be an alternative to human evaluations? In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 15607–15631, 2023. doi: 10.18653/v1/2023.acl-long.870.

[5] DeepSeek-AI, Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, et al. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.

[6] Yanai Elazar, Nora Kassner, Shauli Ravfogel, Abhilasha Ravichander, Eduard Hovy, Hinrich Schütze, and Yoav Goldberg. Measuring and improving consistency in pretrained language models. *Transactions of the Association for Computational Linguistics (TACL)*, 9:1012–1031, 2021.

[7] Alexander R. Fabbri, Wojciech Kryściński, Bryan McCann, Caiming Xiong, Richard Socher, and Dragomir Radev. SummEval: Re-evaluating summarization evaluation. *Transactions of the Association for Computational Linguistics*, 9:391–409, 2021. doi: 10.1162/tacl_a_00373.

[8] Yuanning Feng, Sinan Wang, Zhengxiang Cheng, Yao Wan, and Dongping Chen. Are we on the right way to assessing LLM-as-a-judge?, 2025. URL https://arxiv.org/abs/2512.16041.

[9] Gemini Team, Rohan Anil, Sebastian Borgeaud, Yonghui Wu, Jean-Baptiste Alayrac, et al. Gemini: A family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*, 2023.

[10] Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

[11] Andong Hua, Kenan Tang, Chenhe Gu, Jindong Gu, Eric Wong, and Yao Qin. Flaw or artifact? rethinking prompt sensitivity in evaluating LLMs, 2025. URL https://arxiv.org/abs/2509.01790.

[12] Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, et al. Mistral 7B. *arXiv preprint arXiv:2310.06825*, 2023.

[13] Guneet Kohli. Nine judges, two effective votes: Correlated errors undermine LLM evaluation panels, 2026. URL https://arxiv.org/abs/2605.29800.

[14] Haitao Li, Qian Dong, Junjie Chen, Huixue Su, Yujia Zhou, Qingyao Ai, Ziyi Ye, and Yiqun Liu. LLMs-as-Judges: A comprehensive survey on LLM-based evaluation methods. *arXiv preprint arXiv:2412.05579*, 2024.

[15] Songze Li, Chuokun Xu, Jiaying Wang, Xueluan Gong, Chen Chen, Jirui Zhang, Jun Wang, Kwok-Yan Lam, and Shouling Ji. LLMs cannot reliably judge (yet?): A comprehensive assessment on the robustness of LLM-as-a-judge, 2025. URL https://arxiv.org/abs/2506.09443.

[16] Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narayanan, Yuhuai Wu, Ananya Kumar, et al. Holistic evaluation of language models. *Transactions on Machine Learning Research*, 2023.

[17] Stephanie Lin, Jacob Hilton, and Owain Evans. TruthfulQA: Measuring how models mimic human falsehoods. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*. Association for Computational Linguistics, 2022. doi: 10.18653/v1/2022.acl-long.229.

[18] Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. G-Eval: NLG evaluation using GPT-4 with better human alignment. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 2511–2522, 2023. doi: 10.18653/v1/2023.emnlp-main.153.

[19] Moran Mizrahi, Guy Kaplan, Dan Malkin, Rotem Dror, Dafna Shahaf, and Gabriel Stanovsky. State of what art? A call for multi-prompt LLM evaluation. *Transactions of the Association for Computational Linguistics*, 12:933–949, 2024. doi: 10.1162/tacl_a_00681.

[20] Justin D. Norman, Michael U. Rivera, and D. Alex Hughes. Reliability without validity: A systematic, large-scale evaluation of LLM-as-a-judge models across agreement, consistency, and bias, 2026. URL `https://arxiv.org/abs/2606.19544`.

[21] OpenAI, Josh Achiam, Steven Adler, Sandhini Agarwal, et al. GPT-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.

[22] Amirhossein Razavi, Mitra Soltangheis, Negar Arabzadeh, Sara Salamat, Morteza Zihayat, and Ebrahim Bagheri. Benchmarking prompt sensitivity in large language models. *arXiv preprint arXiv:2502.06065*, 2025.

[23] Angelika Romanou, Mark Ibrahim, Candace Ross, Chantal Shaib, Kerem Oktar, Samuel J. Bell, Anaelia Ovalle, Jesse Dodge, Antoine Bosselut, Koustuv Sinha, and Adina Williams. Brittlebench: Quantifying LLM robustness via prompt sensitivity, 2026. URL `https://arxiv.org/abs/2603.13285`.

[24] Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. Quantifying language models' sensitivity to spurious features in prompt design. In *International Conference on Learning Representations (ICLR)*, 2024.

[25] Lin Shi, Chiyu Ma, Weicheng Liang, Xinyuan Diao, Jinwei Pu, Xiaobo Cao, Wenhua Zhao, Hao Liu, and Furu Shen. Judging the judges: A systematic investigation of position bias in pairwise comparative assessments by LLMs. *arXiv preprint arXiv:2406.07791*, 2024.

[26] Aman Singh Thakur, Kartik Choudhary, Venkat Srinik Ramayapally, Sankaran Vaidyanathan, and Dieuwke Hupkes. Judging the judges: Evaluating alignment and vulnerabilities in LLMs-as-Judges. *arXiv preprint arXiv:2406.12624*, 2024.

[27] Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. BEIR: A heterogeneous benchmark for zero-shot evaluation of information retrieval models. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2021.

[28] Peiyi Wang, Lei Li, Liang Chen, Zefan Cai, Dawei Zhu, Binghuai Lin, Yunbo Cao, Qi Liu, Tianyu Liu, and Zhifang Sui. Large language models are not fair evaluators. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.acl-long.511.

[29] Hui Wei, Shenghua He, Tian Xia, Fei Liu, Andy Wong, Jingyang Lin, and Mei Han. Systematic evaluation of LLM-as-a-Judge in LLM alignment tasks: Explainable metrics and diverse prompt templates. *arXiv preprint arXiv:2408.13006*, 2024.

[30] Shihao Weng, Yang Feng, and Xiaofei Xie. Beyond accuracy: Policy invariance as a reliability test for LLM safety judges, 2026. URL `https://arxiv.org/abs/2605.06161`.

[31] Abel Yagubyan. The coin flip judge? reliability and bias in LLM-as-a-judge evaluation, 2026. URL `https://arxiv.org/abs/2606.13685`.

[32] An Yang, Baosong Yang, Beichen Zhang, et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.

[33] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[34] Jingming Zhuo, Songyang Zhang, Xinyu Fang, Haodong Duan, Dahua Lin, and Kai Chen. ProSA: Assessing and understanding the prompt sensitivity of LLMs. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 1950–1976, 2024.